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Author(s): Xavier Giné and Martin Kanz

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The Economic Effects of a Borrower Bailout: Evidence from an Emerging Market

Xavier Giné

World Bank

Martin Kanz

World Bank

We study the credit market impact and real effects of one of the largest borrower bailouts in history, enacted by the government of India against the backdrop of the 2007–2008 financial crisis. We find that the bailout led to a significant reallocation of credit and greater defaults, but had no offsetting positive effect on productivity, wages, or consumption. Post-program loan performance deteriorates faster in districts with greater program exposure, an effect that is not driven by greater risk-taking of banks. Loan defaults become more sensitive to the electoral cycle after debt relief, suggesting strategic default in response to the bailout. (*JEL* G21, G28, O16, O23)

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Since the Great Depression, economic stimulus programs have been ubiquitous in many economies. In their simplest form, such programs provide direct subsidies or income support. In many other cases, however, stimulus programs are implemented through the credit market, typically in the form of debt restructuring and debt relief programs designed to encourage investment and consumption.

Stimulus programs that operate through the credit market have remained controversial for at least two reasons. First, as in the case of stimulus programs more generally, critics question whether ex post credit market interventions can meaningfully affect real economic activity (Agarwal, Ben-David, et al. 2017; Agarwal, Chomsisengphet, and Hassler 2005). Second,

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while proponents argue that credit market stimulus programs can resolve debt overhang problems, increase productivity, and reduce deadweight losses during economic downturns (Bolton and Rosenthal 2002; Mian, Rao, and Sufi 2013), critics have suggested that credit market interventions are a particularly harmful way of enacting a stimulus, as they change the contracting environment and may give rise to moral hazard. Despite this long-standing debate, there exists surprisingly little empirical evidence on how such interventions affect the credit market and the real economy.

In this paper, we address these questions using a natural experiment arising from one of the largest borrower bailout programs in history, enacted by the government of India against the backdrop of the global financial crisis of 2007–2008. The program, known as the “Agricultural Debt Waiver and Debt Relief Scheme” (ADWDRS), consisted of unconditional debt relief for more than 40 million rural households, and amounted to a volume of U.S.\$12 billion (approximately 1% of GDP).

The main challenge in identifying the causal effect of a stimulus program is the difficulty of constructing a valid counterfactual. In particular, one might worry that estimates of program impact are endogenous to preexisting economic trends, since local exposure to the program is usually a function of economic conditions—in this case loan delinquencies—before the program. To address this concern, we take advantage of the ADWDRS program eligibility rules, which generate plausibly exogenous variation in local exposure to the bailout.

We find that the bailout led to a significant *ex post* reallocation of credit and substantial strategic default in districts with high program exposure. Specifically, a one-standard-deviation increase in the share of credit waived as a result of the bailout leads to a 4–6% decline in the number of new loans and a 14–16% decline in the amount of post-program lending. This reallocation of credit in response to the bailout disproportionately affects regions with higher exposure to the program. While districts with high (above-median) program exposure received an average of 36 cents of new lending for every 1 dollar of debt relief, districts with low (below-median) program exposure received approximately 4 dollars of new lending for every 1 dollar of debt relief. Despite this reallocation of post-program lending to observably less risky districts, we document a significant decline in loan performance in areas with higher bailout exposure. This increase in defaults is concentrated among borrowers that had previously been in good standing. We perform several tests to rule out that defaults are driven by changes in loan sizes or credit market conditions. In addition, the pattern of our results indicates that the increase in defaults after the program cannot be explained by greater bank risk-taking. This suggests that debt relief led to significant borrower moral hazard, and had a particularly negative impact on the credit discipline of borrowers who were not in default at the time the bailout was enacted.

In the second part of our analysis, we examine the effect of debt relief on the real economy. To do so, we use regionally disaggregated data on rural

productivity, unskilled wages, and employment, which are the main outcomes one would expect to be affected if the bailout was indeed effective at resolving debt overhang problems or relaxing borrowing constraints by freeing up pledged land collateral.¹ Our results identify zero or marginally negative effects for each of these outcomes. This is consistent with household-level evidence on the impact of the bailout,² implies a low spending multiplier of debt relief, and suggests that the program did not have the impact on real economic activity suggested by its proponents. Interpreted against the background of the credit market results, these findings suggest that banks reallocated credit away from districts with high bailout exposure, potentially in anticipation of the adverse moral hazard consequences of debt relief. While this improved the efficiency of credit allocation, it counteracted any potentially positive impacts that the bailout might have had on real economic activity, as credit was redirected away from districts where the incentive effects of the bailout would have been the greatest.

There are several reasons that make India's bailout for rural households a particularly attractive setting to explore the impact of a credit market stimulus. First, India's ADWDRS bailout shares many features with debt forgiveness programs that have been enacted in other settings. In the United States, debt moratoria for rural households were widespread in the aftermath of the Great Depression (see Alston and Rucker 1987). More recently, it has been argued that weak household balance sheets precipitated the financial crisis of 2007–2008 (Mian and Sufi 2014; Mian, Rao, and Sufi 2013), and debt relief and restructuring programs have been proposed as a potential solution. Indeed, the available evidence suggests that the impact of the financial crisis was especially severe in regions with higher pre-crisis levels of household debt, and numerous countries, including the United States, have enacted large debt restructuring programs in the years after the crisis to reduce overall levels of household debt in the economy.³ While such programs have been enacted in many settings and at different points in time, debt forgiveness initiatives have generally been a more frequently used policy in developing countries. Similar to the program we

¹ See, for example, Vig (2013) for the impact of a collateral shock on financial contracting and firm-level real outcomes. See Kerr and Nanda 2009 and Schmalz, Sraer, and Thesmar (2017) for more on access to collateral and entrepreneurial activity and Campello and Larrain (2016) on expanded access to collateral, borrowing, and economic activity.

² See Kanz (2016) and De and Tantri (2014) for evidence on individual responses to debt forgiveness.

³ See "Chapter 3: Dealing with household debt" in International Monetary Fund (2012), *World Economic Outlook* for a detailed discussion of household debt relief programs, which includes examples ranging from Scandinavia in the 1990s to Colombia in 1999, and Hungary, Iceland, and the United States after the Great Recession.

Using data for the United States, Mian, Rao, and Sufi (2013) show that higher local levels of household debt in the runup to the 2007–2008 financial crisis led to a more severe contraction of economic activity and a slower recovery once the crisis hit. The same general pattern has been shown to hold in a panel of OECD economies (see International Monetary Fund 2012), and is also broadly consistent with the finding that economy-wide crisis episodes tend to be more severe when they are preceded by a credit boom (see, for example, Claessens, Kose, and Terrones (2012) and Jordà, Schularick, and Taylor (2013)).

study, they have often targeted the rural sector, with the intention of alleviating debt overhang and restoring access to finance. Some recent examples include a U.S.\$2.9 billion bailout for farmers in Thailand and the restructuring of more than U.S.\$10 billion of household debt in Brazil.⁴

Second, India's ADWDRS program was an unanticipated one-off initiative that left the institutional environment unchanged. The program was the first nationwide debt relief initiative in India in nearly two decades. There was no public debate of debt relief in the months preceding the bailout, and the program was enacted as an executive decision that was not debated in parliament, making it very unlikely that borrowers could have anticipated the timing of the program or specific criteria for eligibility.⁵ Importantly, debt relief settlements affected the terms of existing loans, but did not modify the rules, institutions, and regulations governing new lending.

Third, unlike any previous debt relief initiative in India, eligibility for the program depended on the amount of land pledged at the time of loan origination—typically many years prior to the program—with households below a retroactively imposed collateral threshold qualifying for full and unconditional debt relief. This rule provides a source of plausibly exogenous variation in program exposure, because it implies that the share of credit that could qualify for the program is a function of total defaults in a given district as well as the historically determined land distribution, which is exogenous over the time period studied in this paper. The fact that the same collateral-based eligibility cutoff applied uniformly across all districts of India also ensures that our results are not biased by politically motivated targeting of transfers to specific districts or electoral constituencies.

This paper contributes to several strands of the literature. First, our results relate to a literature on the real and financial impact of debt relief and debt restructuring programs. Agarwal, Ben-David, et al. (2017) study a subsidized mortgage restructuring program in the United States and find that debt restructuring led to a modest reduction in foreclosures but had no effect on real outcomes, such as durable and nondurable consumption. Alston (1984) and Alston and Rucker (1987) study debt moratoria in the United States after the Great Depression, and find that the short-term benefits of debt relief may come at the cost of substantial moral hazard and credit rationing in the longer run. Dobbie and Song (2014, 2015) study the real effects of debt relief through loan modifications and personal bankruptcy settlements and find evidence of labor supply and employment effects. More broadly, our results are also related to a literature examining the ability of non-credit-market stimulus programs to affect economic activity (Aaronson, Agarwal, and French 2012; Agarwal and Qian 2015; Mian and Sufi 2012). While our

⁴ See USDA Economic Research Service Briefings, <http://www.ers.usda.gov>.

⁵ We document the absence of anticipation effects in Online Appendix A.

analysis differs from this line of research by focusing specifically on a credit market stimulus, our results on real effects align with the general findings from this literature and suggest a low spending and investment multiplier from debt relief.

Because the ADWDRS bailout entailed the recapitalization of banks, our results are also related to a literature on bank recapitalizations (Hoshi and Kashyap 2010; Philippon and Schnabl 2013) and the transmission of bank liquidity shocks to the real economy (Cetorelli and Strahan 2006; Giannetti and Simonov 2013; Jayaratne and Strahan 1996; Khwaja and Mian 2008; Paravisini, Rappoport, Schnabl, and Wolfenzon 2014; Schnabl 2012). Most closely related to our paper, Breza and Kinnan (2016) study a negative shock to the liquidity of microfinance lenders in India and find that reductions in credit supply are associated with decreases in unskilled daily wages, household wage income, and consumption.

Finally, we contribute to a literature on the political economy of credit (Behn, Haselmann, et al. 2014; Cole 2009). Although a uniform eligibility rule applied across India, so that no strategic local targeting of benefits was possible, there was significant concern that the bailout could aggravate moral hazard by raising expectations of future politically motivated government interventions into the credit market. Consistent with this hypothesis, we find that loan performance deteriorates sharply in districts with greater exposure to the bailout. We also show that defaults are sensitive to the electoral cycle and this pattern is magnified after the bailout. These results provide evidence of strategic default in response to a government intervention into the credit market, and suggest that the anticipation of future credit market interventions is an important channel through which moral hazard in loan repayment is intensified.

1. India's Bailout Program for Rural Households

India's debt relief program for rural households was announced in March 2008, against the backdrop of the global financial crisis of 2007–2008.⁶ The goal of the program was twofold. First, owing to a long history of directed lending to the rural sector, Indian banks had accumulated significant amounts of nonperforming loans. The bailout was intended to strengthen Indian banks by eliminating these nonperforming assets from their books and recapitalizing them in the process. Second, the reduction of household debt as a result of ADWDRS was intended to strengthen household balance sheets and act as a direct stimulus in anticipation of negative demand shocks to India's rural economy in the aftermath of the global economic crisis.⁷

⁶ The bailout occurred at a time when the Indian economy was not experiencing any significant macroeconomic shocks. This helps us isolate the impact of debt relief from contemporaneous economic events that might confound our estimates.

⁷ In 2008, agriculture accounted for approximately 15% of India's GDP and 55% of total employment.

Introduced a year before national elections, the program also represented a large transfer from urban to rural voters and served some more direct political motivations.⁸

There was no public debate of debt relief in the months prior to the bailout, and all available evidence suggests that both the timing of the bailout as well as the specific criteria for program eligibility came as a surprise to those affected.⁹ The program was implemented as an executive decision, which meant that it did not go through the standard legislative process and was not debated in parliament prior to the program announcement. Instead, the program was introduced as part of the Indian federal budget and voted on only after the bailout had already been announced.¹⁰

The rules for program eligibility were kept deliberately simple to expedite the processing of claims and minimize opportunities for leakage and corruption. In contrast to any prior debt relief initiatives, individual eligibility for the bailout depended on the amount of land pledged as collateral at the time a loan was originated, typically several years before the program. Borrowers who had pledged less than two hectares of land were eligible for full debt relief, while those that had pledged more than two hectares qualified for 25% debt relief conditional on repaying the remaining balance. Loans qualified for the program if they (i) had been originated between December 31, 1997, and December 31, 2007, (ii) were more than 90 days past due as of December 31, 2007, and (iii) remained in default until February 28, 2008. Importantly, the eligibility rules, including the collateral cutoff, were applied retroactively, so that there was no scope for manipulation around the program date. In addition, ADWDRS was the first debt relief program in India's history to use landholdings as a basis for eligibility, which makes it unlikely that borrowers could have anticipated the specific criteria required to benefit from the bailout.

The program rules also ensured that bailout benefits could not be targeted locally, for example to favor electorally competitive constituencies. The same two-hectare cutoff was applied everywhere in India and could not be modified by local bureaucrats or politicians. Therefore, the only possibility to target benefits locally would have been to manipulate historical land records. This was made practically impossible by the fact that the eligibility criteria, including the land collateral cutoff, were applied retroactively and

⁸ The fact that the program was enacted one year before national elections does not affect the identification, since all units of observation are on the same electoral cycle with respect to national elections.

⁹ The report of a government committee looking into the issue of rural indebtedness, submitted only months before the bailout, had not included debt forgiveness as one of its recommendations (see Government of India 2007). We present several tests to rule out "news leaks" and anticipation effects in Online Appendix A. These include evidence from the frequency of bailout-related terms in news reports and online searches around the program announcement.

¹⁰ The ruling coalition held a majority in both houses of the Indian parliament at the time, and the program passed without amendments.

there were typically multiple independent records of a borrower's land collateral.¹¹

Implementation of the program began in June 2008. Every bank branch in the country was required to compile and publicly post a full list of debt relief beneficiaries, including details on the loan amount, date of origination, land collateral, and amount eligible to be waived under the program. Unconditional debt relief for borrowers with less than two hectares of land collateral was processed immediately. The deadline for settling claims for loans with more than two hectares of land collateral was extended several times, first to December 2009, and then to December 2010.¹² Participation in the program was near universal, owing to the fact that debt relief was mandatory for qualifying households below the two-hectare cutoff, and a high percentage of eligible households above the cutoff—for whom participation was optional—chose to opt into the program to obtain partial debt relief.¹³ Borrowers who were granted debt relief had their collateral cleared through an official entry on their land documents, and were free to use their collateral to access new loans. Banks were, in principle, required to make bailout beneficiaries eligible for new loans. However, both anecdotal evidence and our results suggest that many banks did not follow this directive and reduced credit supply in regions with high exposure to the bailout.

The program was widely publicized and received significant attention in the national and international news media. Proponents of the bailout heralded it as a cure for endemic problems of debt overhang and poor investment incentives in the Indian rural economy. Responses to the program from the financial sector and many international observers were more cautious and warned, in particular, of the potentially detrimental consequences of the bailout for credit discipline and future access to credit for borrowers that had benefited from the program.¹⁴

2. Data and Descriptive Statistics

We examine the impact of India's debt relief program for rural households, using a panel data set on credit market outcomes and real economic activity over the period 2001 to 2012.¹⁵ The primary unit of analysis is an Indian census district,

¹¹ In Online Appendix B, we test for the politically motivated endogenous targeting of bailout benefits and show that this can be ruled out. There is no indication that politically more contested regions benefited disproportionately from the bailout.

¹² To ensure that we accurately capture the total amount of debt relief, we use data as of December 2011, when the program was closed and all claims had been settled.

¹³ Borrower-level data indicates that more than 70% of households above the cutoff opted into the program. Full debt relief for borrowers below the cutoff accounted for approximately 81% of total debt forgiven.

¹⁴ *The Economist*, for example, noted "Some fear that the government's largesse will do lasting damage to a culture of prudent borrowing, productive investment and prompt repayment." See "Waiving, not drowning: India writes off farm loans. Has it also written off the rural credit culture?" *The Economist*. June 3, 2008.

¹⁵ Between 2001 and 2012, 47 new districts were created, typically by bifurcating existing ones. In our analysis, we aggregate all data to the level of India's 2001 census districts.

an administrative unit roughly comparable to a U.S. county. In the base year 2001, India had 593 census districts with an average population of 1,731,897 inhabitants. Our data set covers 489 districts, which account for approximately 96% of the Indian population and 94% of total bank lending in 2001.¹⁶

The data set contains three main types of information. First, it records cross-sectional exposure to the bailout, measured as the share of total agricultural credit eligible to be written off under the program in each district. Second, it contains detailed information on district-level credit market outcomes, including lending and loan performance. Third, it contains information on real economic outcomes, including agricultural productivity, rural unskilled wages, and per capita consumption expenditure. In this section, we describe each set of variables in turn.

2.1 Measuring local bailout exposure

To measure local exposure to the bailout, we use data on debt relief amounts collected from the records of the State Level Bankers' Committee, the administrative body responsible for recording data on public credit market programs, in each state of India.¹⁷ For each district, we observe the amount of debt relief granted, the amount of credit eligible to be forgiven, and the amount of credit qualifying for conditional and unconditional debt relief, respectively.

We use this information to construct a continuous measure of cross-sectional program exposure. Because participation in the program was mandatory for qualifying borrowers below the two-hectare collateral cutoff and optional for borrowers above the cutoff, our preferred measure of program exposure uses the share of credit eligible to be forgiven under the program to avoid selection issues. Letting $credit_i$ denote the total amount of agricultural credit outstanding in district i at the time the bailout was implemented, we define program exposure as:

$$\text{Bailout share}_i = s_i \left(\frac{credit_{c,i} + .25\bar{\kappa} credit_{-c,i}}{credit_i} \right) \quad (1)$$

where s_i denotes the share of credit in default, c is the share of total debt owed by households below the two-hectare collateral cutoff, $-c$ is the share of debt owed by households above the cutoff, and $\bar{\kappa}_i$ denotes the fraction of borrowers eligible for partial debt relief opting into the program. We let $\bar{\kappa}_i = 1$ for all i ,

¹⁶ Bank lending in India is typically concentrated within a district, and very little lending takes place across district boundaries. This is the result of India's long-standing branch banking regulations, under which branch licenses and lending targets are typically assigned at the subdistrict level.

We exclude branches in India's largest metropolitan areas, Ahmedabad, Bangalore, Chennai, Jaipur, Hyderabad, Kolkata, Mumbai, and Delhi, where this mapping between bank and borrower location is less likely to hold.

¹⁷ We were able to obtain this information for 24 of India's 29 states. Data were not available for the northeastern states Arunachal Pradesh, Assam, Meghalaya, Mizoram, and Nagaland. In the base year of our study, these states account for less than 5% of total credit outstanding.

Table 1
Program exposure

	Bailout share
Mean	0.327
Median	0.286
StDev	0.224
Min	0.002
Max	0.990

This table reports summary statistics for our measure of program exposure. The variable bailout share measures the total amount of credit eligible for the bailout as a share of total agricultural credit outstanding at the time of the program announcement, March 2008, for the 489 districts in the sample. To facilitate the interpretation of coefficients, the variable is normalized to mean zero and standard deviation one in all following tables.

that is, we assume full compliance above the cutoff. This will avoid bias due to the possibility of selective compliance among households eligible for partial debt relief and result in a lower bound estimate of the treatment effects.

Table 1 reports summary statistics for the cross-sectional intensity of program exposure. At the time the program came into effect, the median district in our sample had U.S.\$47 million of agricultural credit outstanding and saw approximately one-third of this amount written off as a result of the program (28.4%, s.d. = 22.4). Figure 1 plots the share of credit eligible to be forgiven in each district of India, and illustrates that there is also substantial geographical variation in program exposure. It is worth noting that program exposure does not appear to correlate significantly with state boundaries or pre-program credit, suggesting that exposure to the bailout is driven primarily by local variation in economic shocks, rather than differences in state institutions or other omitted factors, such as the development of local credit markets.

2.2 Lending and loan performance

2.2.1 Bank lending. The main source of credit market data is the Reserve Bank of India's Basic Statistical Returns of Commercial Banks in India database. This data set is a panel of bank lending at the district level, which covers the lending activities of all commercial banks in India at more than 100,000 bank branches across the country. Unless otherwise indicated, we focus on direct agricultural credit, the type of credit most directly affected by the ADWDRS program, and construct two variables, measuring the total amount of credit, and the total number of loans outstanding for each district-year over the time period 2001 to 2012, respectively. Since the vast majority of rural credit in India is supplied by state banks, which are publicly listed but majority government owned, the loans in this data set are predominantly loans made by state banks.¹⁸ Summary statistics are reported in Table 2. In the base year 2001, the median district in our sample had a total of 22,744 agricultural loans with an average (median) loan size of U.S.\$515 (U.S.\$379) outstanding.

¹⁸ In 2001, the base year of our study, private-sector banks accounted for less than 2.5% of agricultural credit, and less than 5% of total credit in rural India.

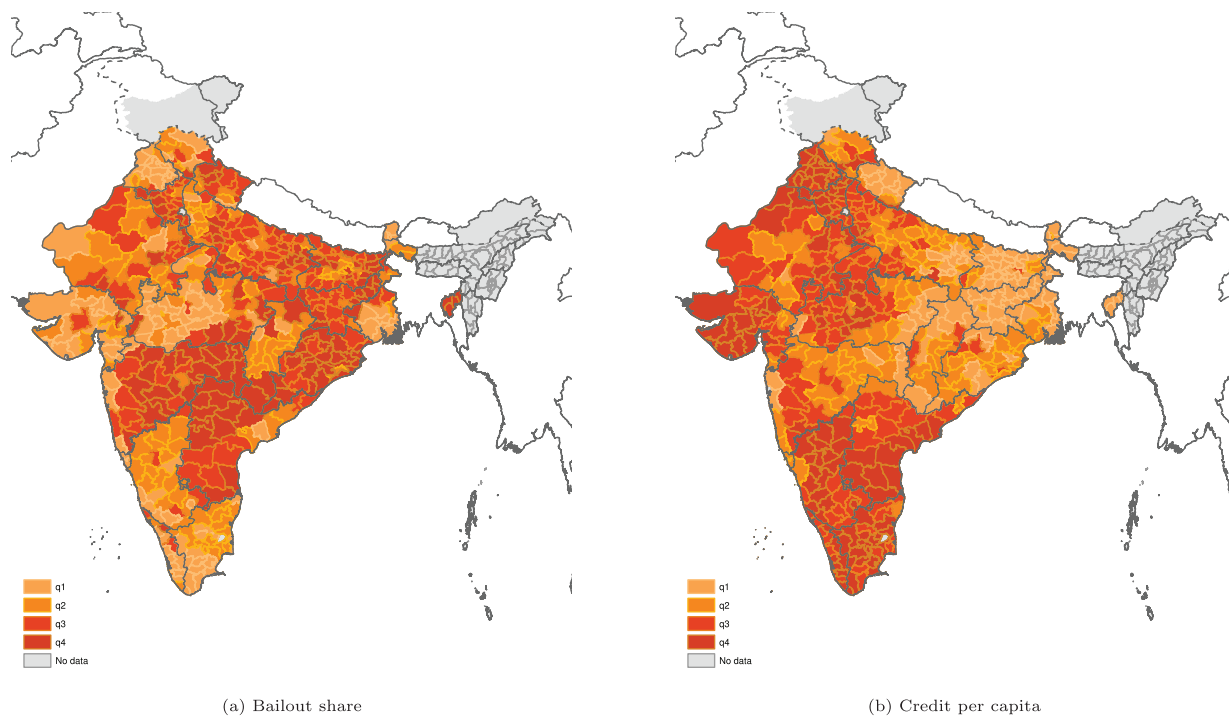


Figure 1
Treatment intensity

Panel (a) plots cross-sectional exposure to the bailout, measured as the share of credit outstanding that was eligible to be forgiven under the program in each district. Panel (b) plots the geographical distribution of agricultural credit per capita as of March 2008, the time of the program announcement. Darker colors indicate higher amounts of credit per capita and bailout exposure, respectively. No debt relief data were available for areas shaded in gray.

Table 2
Summary statistics

	(1)	(2)	(3)	(4)
	N	Mean	Median	StDev
I. Lending and loan performance				
Total number of loans	6,907	10.14	10.41	1.54
Total loan amount	6,907	9.27	9.52	1.80
Nonperforming loans, share of accounts	6,437	0.18	0.10	0.22
Nonperforming loans, share of credit	6,574	0.14	0.06	0.20
Increase in nonperforming share of accounts	5,740	0.42	0.00	0.49
Increase in nonperforming share of credit	5,891	0.44	0.00	0.50
II. Real outcomes				
Productivity per hectare	5,300	0.17	0.14	0.10
Rural unskilled wage	966	4.15	4.09	0.39
Per capita consumption	2,970	11.90	11.79	0.52
III. District characteristics				
Land holdings below 2ha	6,156	0.74	0.79	0.22
Monsoon rain, % of 50-year mean	6,168	94.95	88.64	45.69
Drought years, pre-program	7,008	0.87	0.00	1.49
Total population	6,972	14.00	14.21	1.02
Rural share	6,972	0.78	0.82	0.17
Income per capita	6,432	0.17	0.03	0.48

Total number of loans is the log number of total agricultural loans outstanding. Total loan amount is the log amount of total agricultural credit. Nonperforming loans, share of accounts, and nonperforming loans, share of credit, are the shares of nonperforming loans and nonperforming credit in a given district, respectively. Increase in nonperforming share of accounts is an indicator equal to one for each year in which a district experienced an increase in the share of nonperforming loans. Increase in nonperforming share of credit is an indicator equal to one for each year in which a district experienced an increase in the share of nonperforming credit. Productivity per hectare is the total revenue from all crops produced in a district at 2001 prices, divided by the total area planted. Rural unskilled wage is the log average daily wage for agricultural labor. Per capita consumption is the log of monthly per capita household consumption. Land holdings below two hectares is the share of total recorded landholdings in a district that are smaller than two hectares. Monsoon rain is the average precipitation recorded between June and September as a share of the district's 50-year rain average over the same period. Drought years is the number of years between 2001 and 2007 in which monsoon rainfall in a district was less than 75% of its long-run average. Total population is the log of a district's population. Rural share is the share of a district's population living in rural areas. Income per capita is a district's 2001 log agricultural revenue per capita.

We augment our district-level data with information on lending and loan performance at the bank-district level. Since data at this level of disaggregation is not available from public databases, we construct a new data set using hand-collected proprietary data from India's four largest commercial banks.¹⁹ This data set contains information on lending and loan performance for 1,783 bank-district pairs observed annually between 2005 and 2012. In the base year, the data cover lending at 27,678 bank branches that account for approximately 40% of the Indian branch network and 62% of total credit. We again focus on agricultural credit and construct bank-district-level measures of the amount and number of loans outstanding. We use data at this lower level of aggregation for a series of tests, which ensure the robustness of our results to unobserved credit market shocks at the district level.²⁰

2.2.2 Loan performance. Because information on loan performance at the subnational level is not disclosed by the regulator, the bank-district panel is

¹⁹ All banks in the bank-district sample are public-sector banks.

²⁰ These tests are reported in Section 5 and Online Appendix E.

also our main source of information on loan performance. Using this data set, we construct variables measuring the share of nonperforming credit, and the share of nonperforming loans. To do so, we use the standard definition, which classifies a loan as nonperforming if it remains more than 90 days past due. The median bank branch in our sample reports 6% (s.d.=20) of loans as nonperforming. In the pre-program period, this figure stands at 13.6% (s.d.=12.3), and drops to 6.25% (s.d.=7.3) as a result of the bailout.

2.3 Productivity, wages, and consumption

2.3.1 Productivity. Our primary source for data on agricultural productivity is the Indian Agricultural Statistics Database, provided by the Indian Ministry of Agriculture, which contains district-level information on crop yields and area planted for the most common crops grown in India. We combine these data with commodity prices to calculate our local productivity variable, which measures the value of agricultural production per hectare of cultivated land for each district and year in the sample. Because we want to focus on the component of productivity that can be affected by households, and is unaffected by exogenous price fluctuations, we use constant commodity prices for the base year 2001, so that our measure of productivity may be interpreted as a measure of quantity total factor productivity (TFPQ) that is unaffected by variation in commodity prices.

2.3.2 Real wage. The Agricultural Statistics Database also provides district-level information on wages for unskilled rural labor. Wages are reported seasonally, for every district in India and for a range of agricultural occupations. We focus on unskilled daily wages for field and non-field agricultural labor and calculate a district's real wage as the average of these two wage groups, normalized with respect to the base year of our study. We average wages over all reported crop seasons to account for seasonal variation in labor demand.

2.3.3 Consumption expenditure. Data on per capita consumption expenditure are calculated from the Indian National Sample Survey (NSS). The NSS is conducted annually as a repeated cross-sectional survey of approximately 75,000 rural and 45,000 urban households. We use data from four survey rounds available for the time period covered by our data set—two before and two after the bailout—and calculate monthly per capita consumption expenditure following the standard definition used in the literature. Since the survey is representative at the level of the survey unit, rather than the district (a survey unit typically consists of several census districts), we restrict the sample to districts with at least 100 survey households.

2.4 Additional data sources

2.4.1 Weather shocks. We measure weather shocks using local variation in monsoon rainfall. Rainfall data are taken from the Indian Meteorological

Department's climatological database and measure total monthly precipitation for each district. Based on these data, we construct a variable measuring total monsoon precipitation between July and September for each year as a percentage of a district's long-run rainfall average over the same period.²¹ Based on the same data set, we additionally construct a dummy variable indicating years in which a district experienced a drought shock. This variable takes on a value of one for any year in which a district's total monsoon precipitation between the months of July and September was below 75% of the district's long-run average. Note that we measure both drought years and monsoon precipitation relative to a district's long-run rainfall average to avoid a mechanical correlation of the variable with fixed district characteristics and economic outcomes at the district level.

2.4.2 State elections. Previous work has shown that lending of state banks in India closely tracks the electoral cycle.²² Hence, we control for the time until the next state election in each district, using data on the timing of state elections from the Election Commission of India. State elections occur every five years and are staggered across states, with an average of about four Indian states holding state assembly elections in any given year in our data. Because state governments have the power to call early elections, we use the distance to the next scheduled, rather than actual, election to avoid endogeneity in the timing of elections.

2.4.3 District characteristics. Finally, our data set contains information on additional district characteristics from the Indian Census and Agriculture Census. These include the population and land distribution of each district. Additional details on data sources and the definition of variables are available in Online Appendix F.

3. The Credit Market Impact of the Bailout

3.1 Empirical strategy

This section describes the identification strategy we use to estimate the impact of the bailout on credit market outcomes. We take advantage of the fact that the rules for program eligibility introduce plausibly exogenous geographical variation in the share of debt forgiven at the local level.

The program rules specified that a household qualified for debt relief if it was in default at the time of the program date and had pledged less than two hectares of land collateral. The share of outstanding debt eligible to be bailed out in a given locality is therefore jointly determined by the fraction of borrowers in

²¹ We use 50-year averages, also available from the Indian Meteorological Department.

²² Cole (2009) shows that lending by Indian government banks increases 5–10 percentage points in election years.

default and the fraction of defaulters below the two-hectare collateral cutoff.²³ In the cross-section, both variables are determined exogenously by the land distribution and the time series of economic shocks experienced by borrowers ahead of the bailout, respectively. This, in turn, generates plausibly exogenous variation in the share of debt eligible to be bailed out in a given district and allows us to estimate the impact of the bailout using a difference-in-differences approach that compares outcomes across districts with different levels of program exposure before and after the date the bailout was implemented.

3.2 Difference-in-differences estimation

We estimate the impact of the bailout at the district level, using basic difference-in-difference regressions of the form:

$$\ln(C_{it}) = \alpha_i + \beta_t + \gamma E_{it} + X'_{it}\zeta + \varepsilon_{it} \quad (2)$$

where C_{it} is a credit market outcome for district i in year t . The variable $E_{it} = \text{bailoutshare}_i * \text{post}_t$ measures a district's exposure to the program, α_i are district fixed effects that absorb all time-invariant district characteristics, β_t are time fixed effects, and X_{it} is a vector of controls, which includes monsoon rainfall as a percentage of its long-run average.²⁴ The coefficient of interest γ measures the effect of program exposure in terms of year-on-year changes, and the error term ε_{it} captures any omitted factors, including deviations from linearity. We drop the year in which the program came into effect, to rule out any mechanical correlation between treatment intensity and outcomes of interest.

3.2.1 Instrumenting program exposure. To further ensure that our results are not driven by endogeneity of the treatment with respect to preexisting economic trends, we benchmark our difference-in-differences results against estimates from instrumental variables regressions based on the program eligibility rules. Specifically, we use the feature that a loan had to be both in default and backed by land collateral of less than two hectares to qualify for full debt relief. This generates two sources of exogenous variation that allow us to construct an instrument for program exposure.

The first source of variation is the time series of weather shocks experienced by a district in the five years prior to the program announcement, which is a strong predictor of agricultural loan defaults. We use as our measure of

²³ While the cutoff was chosen to benefit the median voter, it is important to note that the identification does not rely on the precise location of the cutoff, since variation in bailout exposure is the result of exogenous shocks to the distribution of defaulters ahead of the program that vary geographically. The specific cutoff that is chosen therefore affects the overall amount of bailout benefits but leaves the source of variation unaffected. The only requirement for this approach to be valid is that the same cutoff be applied uniformly throughout the country, so that no endogenous local targeting of bailout benefits is possible. We show that this was the case in Online Appendix B.

²⁴ Note that credit market and real outcomes are measured in March, the end of the Indian fiscal year. The monsoon precipitation variable measures rainfall for the monsoon season ending in September of the same fiscal year.

weather shocks the number of drought years experienced by a district in the five-year period before the program, and define a drought year as a year in which total monsoon rain was below 75% of the district's long-run average monsoon precipitation.

The second source of variation is a district's historical land distribution, which determines the fraction of households in default that fall below the two-hectare cutoff and are thus eligible for full debt relief. The present-day land distribution of India's districts is largely the result of land reforms enacted under colonial rule and therefore also exogenous over the time period we study.²⁵ Note that even if one of these two exogenous sources of variation were to affect outcomes directly, through a channel other than program exposure, the interaction between weather shocks and the collateral cutoff becomes relevant only as a result of the program rules. We verify empirically that this is the case, and provide additional identification tests in Section 5.

We proceed in two steps to construct our instrument for program exposure. In the first step, we use the number of drought years, w_{it} , experienced by a district in the period prior to the bailout, interacted with the share of households below the eligibility cutoff, ℓ_i , to obtain a cross-sectional predictor of the share of credit bailed out in each district:

$$Z_i = \sum_{t=1}^{\bar{t}} w_{it} \cdot \ell_i \quad (3)$$

In the second step, we interact this variable with an indicator marking the beginning of the program, to construct a time-varying instrument for program exposure:

$$Z_{it} = \left(\sum_{t=1}^{\bar{t}} w_{it} \cdot \ell_i \right) \times Post_t \quad (4)$$

This yields the first-stage specification

$$E_{it} = \delta_i + \theta_t + \gamma^{fs} Z_{it} + X'_{it} \zeta^{fs} + \epsilon_{it} \quad (5)$$

where, as in Equation (2) above, E_{it} is the program exposure of district i at time t , and X_{it} is a matrix of controls, and we additionally control for year and district fixed effects. We present estimates of the first stage in the Online Appendix. According to our preferred specification in Column (4), the first-stage coefficient γ^{fs} is positive and statistically significant at the 1% level ($\hat{\gamma}^{fs} = 0.241$, s.e. = 0.047), indicating that districts with greater exposure to pre-program drought shocks and more households below the eligibility cutoff indeed saw a significantly higher share of credit bailed out as a result of the program. The F -statistic indicates a strong first-stage relationship.

²⁵ See Banerjee and Iyer (2005) for a discussion of the history and impact of colonial land tenure systems in India.

3.2.2 Exclusion restriction. This identification approach requires that the instrument affects outcomes only through its effect on bailout exposure. While this assumption cannot be tested directly, in Section 6 we perform a number of falsification exercises to assess its plausibility. Specifically, we show that there is no correlation between the instrument and any of the outcomes of interest prior to the program, that the instrument does not correlate with district characteristics at the time of the bailout was implemented, and that the instrumental variable results are also robust to the inclusion of interaction terms between district characteristics and the post-program dummy. Taken together, this suggests that the instrument is valid and does not violate the exclusion restriction.

3.2.3 Separating credit supply and credit demand. One remaining concern is the possibility that the treatment effects $\hat{\gamma}$, estimated using Equation (2) could be biased because they confound the impact of the bailout on credit supply with contemporaneous shocks to credit demand. To rule out this possibility, we benchmark our main results against those of an alternative specification that uses credit market data at the bank-district level and absorbs any contemporaneous credit market shocks through the inclusion of bank-district fixed effects. The results of this exercise, which we report in Section 5, suggest that our estimates are not affected by this potential source of bias.

3.3 Impact on credit allocation

In this section, we estimate the effect of the bailout on credit market outcomes. We begin by estimating the impact of debt relief on post-program lending across districts with different levels of bailout exposure, and then present evidence to corroborate the hypothesis that the patterns we observe are driven by changes in the supply of credit that occur as a result of the bailout.

Table 3 reports the results on credit allocation, using the specification described in Equation (2). The dependent variable in Columns (1) to (4) is the log number of loans. The dependent variable in Columns (5) to (8) is the log amount of credit. We observe for each year the total number of loans and total amount of credit outstanding, so that the difference-in-differences coefficient measures the net change in bank lending. To ensure that we can rule out a mechanical impact of the program on our estimates, the year in which the bailout was enacted, and loans were cleared from banks' books, is omitted. For each dependent variable, the first two columns report OLS estimates, and the second set of columns present two-stage least squares (2SLS) results using the instrumental variable strategy described above. In addition to time fixed effects, all regressions control for district fixed effects that absorb any time-invariant district characteristics. Since the share of credit eligible for the bailout is a function of loan delinquencies before the program, one may be concerned that program exposure is confounded with economic trends at the district level. Therefore, our preferred specification throughout the analysis is

Table 3
Credit supply: Effect of the bailout on post-program lending

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Log(number of loans)				Log(loan amount)			
	OLS	OLS	2SLS	2SLS	OLS	OLS	2SLS	2SLS
Bailout share*post	-0.022 [0.014]	-0.017 [0.014]	-0.023 [0.038]	-0.040 [0.039]	-0.034*** [0.013]	-0.031** [0.013]	-0.182*** [0.046]	-0.246*** [0.050]
Observations	4,553	4,553	4,096	4,096	4,553	4,553	4,096	4,096
# of clusters	489	489	433	433	489	489	433	433
R ²	0.735	0.765	0.734	0.759	0.930	0.935	0.919	0.916
District FE	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
State*electoral cycle FE	N	Y	N	Y	N	Y	N	Y

The regressions in this table estimate the impact of debt relief on post-program credit supply. Each column reports results from a separate regression. The dependent variable in Columns (1)–(4) is the log number of loans outstanding. The dependent variable in Columns (5)–(8) is the log amount of credit outstanding. In addition to the fixed effects reported in the table, all regressions control for the deviation of monsoon rainfall from its long run average. Standard errors, in brackets, are heteroscedasticity-robust and clustered at the district level. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

the 2SLS estimate with the full set of controls, which addresses the potential endogeneity of program exposure.

We find that bank lending becomes more conservative as a result of the bailout. While overall lending increases throughout the time period we study, the results indicate that banks reallocate credit away from districts with greater program exposure. This unambiguously improves the overall efficiency of credit allocation, as new lending is being reallocated toward regions that are observably less risky and more productive at baseline.²⁶ The magnitude of credit reallocation in response to the bailout is economically large. Estimates from our preferred specification, reported in Table 3, Columns (4) and (8), indicate that a one-standard-deviation increase in the share of credit covered by the program leads to a 4% decrease in new loans made after the bailout ($\hat{\gamma} = -0.040$, s.e. = 0.039). The shift of bank lending away from high bailout districts is both larger and more precisely estimated when the amount of credit is used as the dependent variable. Our results indicate that a one-standard-deviation increase in program exposure leads to a decrease in new lending of as much as 25% after the program ($\hat{\gamma} = -0.246$, s.e. = 0.050).

To further explore the pattern of reallocation documented in Table 3, we also estimate a modified version of our baseline specification that uses the total amount of credit as the outcome of interest and replaces our measure of program exposure by the total amount of credit eligible for the program in a given district. While this specification does not allow us to fully address potential endogeneity, it has the attractive feature that the difference-in-differences estimate $\hat{\gamma}$ can be

²⁶ In Online Appendix E, Table OA.VI, we report heterogeneous effects by initial district income and productivity. The results highlight that credit is reallocated toward districts with higher initial per capita income and agricultural productivity.

interpreted as the approximate amount of new lending per dollar of bailout exposure. The results, reported in Online Appendix C, suggest an asymmetric reallocation of credit away from high-bailout districts as a result of the program. While districts with high (above-median) program exposure receive only 24 cents of new lending for every 1 dollar of credit written off under the program ($\hat{\gamma}=0.224$, s.e.=0.134), districts with low (below median) bailout exposure receive approximately 2.5 dollars of new lending for every 1 dollar of debt relief ($\hat{\gamma}=2.488$, s.e.=0.284).

The finding that banks channel new lending to observably less risky districts after the bailout may seem somewhat counterintuitive, given that the empirical literature on bank recapitalizations has generally found a positive effect on bank risk-taking. The pattern of reallocation we find is, however, consistent with the hypothesis that the bailout affected incentives for “evergreening”, that is, to keep lending to borrowers close to default in an attempt to avoid marking these loans as nonperforming (see, Peek and Rosengren 2005). Reflecting a long history of directed lending, all banks in India are required to allocate 40% of their net credit to so-called priority sectors, which include agriculture and small-scale industry. While this mandate forces the allocation of a significant share of credit to subprime borrowers, local branch managers also face strong penalties for realizing credit losses.²⁷ This creates strong incentives to keep lending to borrowers close to default. The bailout removed this incentive distortion, as it allowed bank branches to eliminate delinquent borrowers from their books at no financial or reputational cost. Consistent with this hypothesis, we find evidence of a shift in lending away from districts with a higher pre-bailout share of defaulters.²⁸

3.4 Impact on loan performance and moral hazard

We next examine the effect of the bailout on ex post loan performance. Table 4 presents the results, using the same district-level specifications as in Equation (2). The dependent variable in Columns (1) to (4) is a dummy equal to one if the branches in a given district experienced an increase in the number of nonperforming loans between year $t-1$ and t , while the dependent variable in Columns (5) to (8) is a dummy equal to one if bank branches located in a given district experienced an increase in the share of nonperforming credit over the same time period. We again omit all observations for the year in which the bailout came into effect to rule out a mechanical correlation between the program and subsequent changes in loan performance. We find evidence of a strong negative effect of the bailout on loan performance.

²⁷ See Banerjee, Cole, and Duflo (2009) for more on incentives and lending at Indian banks.

²⁸ It is possible that the bailout acted as a substitute for other forms of redistribution, such as political lending. We test this possibility by comparing the magnitude of political cycles in lending before and after the bailout and find no evidence to support this hypothesis. Exposure to the bailout does not reduce political cycles in lending.

Table 4
Moral hazard: Effect of the bailout on post-program loan performance

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Nonperforming loans (number of loans)				Nonperforming loans (loan amount)			
	OLS	OLS	2SLS	2SLS	OLS	OLS	2SLS	2SLS
Bailout share*post	0.034* [0.019]	0.051** [0.021]	0.641*** [0.226]	0.519*** [0.179]	0.070*** [0.019]	0.052*** [0.019]	0.304 [0.186]	0.283* [0.161]
Observations	2,661	2,661	2,398	2,398	2,690	2,690	2,427	2,427
# of clusters	460	460	412	412	460	460	412	412
R ²	0.143	0.215	0.018	0.016	0.147	0.236	0.103	0.183
District FE	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
State*electoral cycle FE	N	Y	N	Y	N	Y	N	Y

The regressions in this table estimate the effect of debt relief on loan performance. Each column reports results from a separate regression. The dependent variable in Columns (1)–(4) is a dummy variable equal to one if a district experienced an increase in the share of nonperforming loans. The dependent variable in Columns (5)–(8) is a dummy variable equal to one if a district experienced an increase in the share of nonperforming credit. In addition to the fixed effects indicated in the table, all regressions control for monsoon rainfall as a percentage of its long run average. Standard errors, reported in brackets, are heteroscedasticity-robust and clustered at the district level. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

The coefficient estimate from our preferred specification indicates that a one-standard-deviation increase in bailout exposure leads to a 52% increase in the probability that a district experiences an increase in the share of nonperforming loans ($\hat{\gamma}=0.519$, s.e.=0.179), and a 28% increase in the probability that a district experiences an increase in the share of nonperforming credit ($\hat{\gamma}=0.283$, s.e.=0.161).

These results allow us to distinguish between bank and borrower risk-taking and indicate that the increase in defaults is concentrated among borrowers that were previously in good standing. First, as we have documented in the previous section, banks reallocate new lending to observably less risky and more productive districts. Hence, the increase in defaults after the program cannot be explained by greater bank risk-taking in response to the bailout. Second, note that the decline in loan performance we observe occurs after loans that were in default have already been written off and thus cleared from banks’ books. Given that Columns (1) to (4) in Table 3 show no evidence of additional lending in high bailout districts, we can also infer that the negative effect of the bailout on loan performance is concentrated among borrowers that were previously in good standing but defaulted in the months after the bailout. This negative externality of the bailout is likely to be aggravated by the fact that the bailout covered only defaulters, while borrowers in good standing received no benefit under the program.

To better understand the mechanism through which the bailout affects defaults among nonbeneficiaries, we perform an additional set of tests. Perhaps the most plausible channel through which the program may affect repayment discipline is the possibility that it generates expectations of similar credit market interventions in the future. To test this hypothesis, we exploit the exogenous

timing of Indian state elections. Existing research shows that rural credit and defaults in India are correlated with the local electoral cycle, so that it would be natural to expect a greater likelihood of political interventions into the credit market ahead of state elections.²⁹ This suggests that one test for strategic default generated by the anticipation of future credit market interventions is to compare the magnitude of default cycles around elections before and after the bailout. We report this test in Online Appendix E. The results confirm that the default cycle around state elections—which is present prior to the bailout—is magnified in the period after the program, consistent with the expectation of more lenient enforcement or future credit market interventions.

Taken together, these results suggest that the bailout had a significant negative effect on the repayment morale of nonbeneficiaries, and that the anticipation of future credit market interventions is an important channel through which incentives for strategic default are intensified.

4. Real Effects of the Bailout

In this section, we explore the effect of debt relief on the real economy. The stated goal of the bailout was to stimulate investment and consumption in anticipation of an economic downturn, and there are at least two channels that would predict a positive impact of the program on real economic activity. First, the bailout generated a large positive collateral shock that should have dramatically relaxed credit constraints, since all beneficiaries were returned the title deeds to the land pledged as collateral at the time their debt was forgiven. Second, to the extent that households were suffering from debt overhang prior to the program, the cancellation of outstanding debts should have resolved any incentive problems that may have constrained productive investment before the bailout.

To explore the impact of the bailout on economic activity, we use an approach similar to that in our analysis of credit market outcomes. Specifically, we estimate the elasticity of nonfinancial outcomes with respect to program exposure, using difference-in-difference regressions of the form:

$$\ln(R_{it}) = \alpha_i + \beta_t + \omega E_{it} + X'_{it}\zeta + v_{it} \quad (6)$$

where R_{it} is an outcome of interest for district i observed at time t , α_i are district fixed effects that absorb all time-invariant district characteristics, β_t are time fixed effects, and X_{it} is a vector of time-varying controls. As before, the variable E_{it} measures a district's exposure to the bailout as the share of credit eligible to be forgiven interacted with the post-program dummy so that the coefficient of interest ω measures the elasticity of real outcomes with respect to program exposure. We estimate this equation using data at the district level, which is the

²⁹ In our sample, defaults in election years increase by approximately five percentage points over the post-program sample mean of 7.25% ($\hat{\gamma} = 0.131$, s.e. = 0.076).

most granular level at which information on real outcomes is available. As in the previous section, we compare the OLS results against estimates from 2SLS regressions, in which bailout exposure is instrumented using the interaction between the number of drought years prior to the bailout and the share of households below the collateral threshold. We again omit the year in which the bailout was enacted to avoid mechanical correlations and make the estimates comparable with the credit market regressions.

We first explore the effect of debt relief on agricultural productivity—the main outcome one would expect to be affected if the bailout was effective at alleviating problems of debt overhang. Debt levels among rural households in India are indeed high, and producers rely heavily on external financing to undertake productive investments. Bailout proponents therefore argued that debt relief would resolve debt overhang, free up land collateral, and enable rural households to engage in productive investment. If the program had this effect, we would expect this to be reflected in greater investment and increased agricultural productivity. To test whether this was the case, we take advantage of a district-level panel on crop yields and estimate treatment effects of bailout exposure on local agricultural productivity, measured as total revenue per hectare cultivated at constant base-period prices. Table 5, Columns (1) to (4), reports the results and shows that the program had no discernible effect on productivity. Estimates from our preferred OLS and 2SLS specifications identify a precise zero ($\hat{\gamma}=0.002$, s.e.=0.002, and $\hat{\gamma}=0.005$, s.e.=0.008, respectively), suggesting that the credit market stimulus did not create investment incentives of a magnitude sufficient to improve productivity.

We next investigate the effect of debt relief on real wages, to test for a possible labor market impact of the bailout. Unskilled labor is an important input to agricultural production. Hence, if the bailout affected households' ability to invest in inputs to production, one might expect to find this effect reflected in rural employment. In Table 5, Columns (5) to (8), we test for an impact of the bailout on rural wages, and again do not find a discernible effect of the bailout. The point estimates are close to zero, marginally negative in the preferred specification ($\hat{\gamma}=-0.052$, s.e.=0.273), and not statistically significant at conventional levels.

Finally, we turn to the effect of the bailout on per capita household consumption expenditure. If households perceived the bailout as a permanent change in their access to credit (and ability to finance productive inputs), one would expect to see an increase in consumption expenditure. We estimate the impact of debt relief on consumption in Table 5, Columns (9) to (12), and find that the bailout had no positive effect on per capita consumption expenditure.³⁰

³⁰ In additional results, reported in Online Appendix E, Table OA. VII, we estimate heterogeneous effects on real outcomes by initial district income and productivity. Despite the fact that the program was designed to favor low-income borrowers, the results show that the program targeting was too coarse to have a positive impact on low-income borrowers and regions. If anything, the results indicate that poorer regions were more adversely affected by the reallocation of credit resulting from the bailout.

Table 5
Real effects of the bailout: Productivity, wages, and employment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Log(productivity)				Log(wage)				Log(consumption)			
	OLS	OLS	2SLS	2SLS	OLS	OLS	2SLS	2SLS	OLS	OLS	2SLS	2SLS
Bailout share*post	−0.000 [0.002]	0.002 [0.002]	0.007 [0.008]	0.005 [0.008]	0.007 [0.026]	0.009 [0.030]	−0.074 [0.124]	−0.052 [0.273]	0.009 [0.009]	0.007 [0.009]	−0.576 [0.364]	−0.621 [0.383]
Observations	2,685	2,685	2,482	2,482	444	444	408	408	1,285	1,285	1,172	1,172
# clusters	486	486	430	430	208	208	187	187	425	425	385	385
R ²	0.239	0.372	0.246	0.376	0.231	0.321	0.086	0.310	0.94	0.947	0.540	0.520
District FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
State*Electoral cycle FE	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y

This table estimate the effect of debt relief on real outcomes at the district level. Each column reports results from a separate regression. The dependent variable in Columns (1)–(4) is agricultural productivity, calculated as annual agricultural revenue per hectare at constant 2001 commodity prices. The dependent variable in Columns (5)–(8) is the average district-level real wage for unskilled agricultural occupations. The dependent variable in Columns (9)–(12) is the district average of monthly per capita consumption expenditure. In addition to the fixed effects indicated in the table, all regressions control for monsoon rainfall as a percentage of its long-run average. Standard errors errors, in brackets, are heteroscedasticity-robust and clustered at the district level. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

In summary, while the policy maker believed that a recapitalization of banks would encourage new lending and stimulate new investment, we find no improvements in agricultural productivity, rural wages, or household consumption—the outcomes one would expect to be most directly affected if the bailout had been effective in addressing debt overhang and strengthening household balance sheets. It is worth noting that this is not evidence that the program had no impact on the incentive and collateral channels. However, the potentially positive effects of debt relief were not transmitted to the real economy, as lenders reallocated credit away from high-bailout districts at the same time that debt forgiveness would have strengthened investment incentives in these regions.

5. Robustness and Identification Tests

5.1 Internal validity

In this subsection, we present additional tests to address potential threats to the internal validity of our study. We first discuss the features of the program that make it unlikely that borrowers could have anticipated the program, and present evidence to rule out news leaks and anticipation effects. We then report tests of the parallel trends assumption, and demonstrate that the results are robust to contemporaneous local credit market shocks as well as potential heterogeneity in the response to economic shocks. Finally, we present a number of placebo exercises to assess the plausibility of the exclusion restriction for the instrument used in our preferred identification strategy.

5.1.1 “News leaks” and anticipation effects. Several features of the program make it unlikely that borrowers could have anticipated the program and acted strategically in ways that would affect the identification. First, debt relief is a relatively uncommon policy in the Indian credit market. The ADWDRS program was the first nationwide debt relief initiative in nearly two decades and the first program ever to make beneficiary status contingent on land collateral.³¹ Second, there was no public debate on issues of debt relief in the policy circles or the media ahead of the program. The report of a government committee looking into rural indebtedness, submitted only a few months before the program (see Government of India 2007) did not mention debt forgiveness as one of its policy recommendations and—as we show in Online Appendix A—terms related to debt forgiveness did not feature in online searches and news reports in the months before the program, but spiked once it was announced. Third, the program rules applied retroactively, so that there was no time between the

³¹ The last nationwide debt relief initiative in India occurred in 1990 (Sud 2009). This program, known as the “Agriculture and Rural Debt Relief Scheme” was based on loan size, rather than land collateral. It benefited approximately 44 million borrowers, but was much smaller in volume, as it only waived up to Rs 10,000 (U.S.\$145).

program announcement and implementation for borrowers to manipulate their landholdings or default status.³² Account-level data confirms that there was no increase in loan originations in the year prior to the program and no relative increase in the share of loans qualifying for a full bailout, which one would expect to see if there had been manipulation around the program threshold.³³

5.1.2 Parallel trends. Identification using a difference-in-differences estimator relies on the assumption that outcomes for the treatment and control groups (in our case, districts with different levels of bailout exposure) follow parallel trends before the program, and would have followed parallel trends in the absence of the intervention. Since our treatment variable is continuous, simple graphical tests may mask deviations from the parallel trends assumption among a subset of districts. We therefore test for bias due to preexisting trends in outcomes of interest, using two placebo experiments. In the first placebo experiment, we move the timing of the bailout from 2008 to a hypothetical program date in 2007. The results, reported in Table 6, Panel A.1, show that the treatment coefficients for the hypothetical program date are close to zero and, in several cases, have a different sign than the baseline results, indicating that the treatment effect at the actual program date is not driven by unobserved time trends. The second placebo experiment, reported in Table 6, Panel A.2, randomly reassigns treatment levels among the cross-section of districts in our sample. We do this using a bootstrap procedure, in which we draw $N = 1,000$ random assignments of our treatment vector, estimate treatment coefficients, and report bootstrap coefficients and standard errors obtained from this exercise. The estimated treatment effects are again close to zero and statistically insignificant, making it unlikely that our results are biased due to deviations from the parallel trends assumption.³⁴

5.1.3 Robustness to contemporaneous credit market shocks. Another potential concern with the credit market results reported in Section 4 is that the lending and loan repayment response to the program could be confounded by contemporaneous credit market shocks. To address this possibility, we benchmark our baseline credit market results against estimates from an alternative identification strategy that uses credit market data at the bank-district level and exploits within-district variation in bailout exposure. To do so, we

³² Not surprisingly for a program of this scale, some irregularities in implementation did occur. Importantly, however, an audit of the ADWDRS program (Government of India 2013) finds instances of program benefits being erroneously granted to ineligible borrowers and benefits being erroneously denied to ineligible borrowers in approximately the same magnitude. This suggests that errors were not the result of large-scale manipulation of eligibility status, in which case we would expect to see deviations in only one direction. It also suggests that we can rule out bias due to nonclassical measurement error.

³³ See Kanz (2016) for evidence from account data and beneficiary surveys.

³⁴ We also verify that the estimates are robust to the inclusion of linear trends at the district level. The results of this exercise are available on request.

Table 6
Robustness: Placebo experiments

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Log(number of loans)	Log(loan amount)	NPL(number of loans)	NPL(loan amount)	Log(productivity)	Log(wage)	Log(consumption)
	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS
Panel A.1: Placebo treatment effects [post=1($t \geq 2007$)]							
Bailout share*post	-0.021	-0.096	-0.116	-0.266	-0.018	0.459	-0.008
95% confidence interval	[0.122 0.080]	[-0.219 0.027]	[0.467 0.234]	[-0.599 0.066]	[-0.061 0.026]	[-1.110 2.027]	[-0.173 0.156]
Observations	2,450	2,450	2,113	2,114	1,725	90	540
R ²	0.753	0.911	0.191	0.270	0.395	0.110	0.981
Panel A.2: Placebo treatment effects [reassigned treatment levels]							
Bailout share*post	-0.011	-0.020	0.027	0.086	0.000	0.020	-0.017
95% confidence interval	[0.435 -0.457]	[1.080 -1.120]	[2.554 -2.501]	[3.669 -3.497]	[0.080 -0.080]	[2.146 -2.105]	[1.823 -1.857]
Observations	4,096	4,096	2,427	2,398	2,482	4,08	1,172
R ²	0.666	0.870	0.188	0.289	0.400	0.120	0.968
District FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
State*Electoral cycle FE	Y	Y	Y	Y	Y	Y	Y

This table reports the results of two placebo experiments that test the parallel trends assumption. The first placebo experiment, in Panel A.1, restricts the sample to the period prior to the program and estimates treatment effects for a counterfactual program date. In the second placebo experiment, reported in Panel A.2, we generate $N=1,000$ vectors containing randomly reassigned values of program exposure and use these placebo treatment vectors to estimate counterfactual treatment effects and bootstrap standard errors. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

derive an estimating equation at the bank-district level from a simple model of credit supply and demand and estimate this equation using data on lending and loan performance at the bank-district level.³⁵

This estimation strategy has two advantages. First, taking the identification from the district to the bank-district level allows us control for district time fixed effects, which absorb any unobserved shocks at the district level. Second, we can identify the bank lending channel separately from the effect that the bailout has on the within-bank reallocation of credit across districts with differential program exposure. While this exercise provides a useful robustness check to verify the direction of post-program loan defaults and credit allocation, we emphasize that the point estimates are not directly comparable with those from our baseline specification, since the structural derivation of the bank-district estimating equation calls for the inclusion of a different set of explanatory variables. Specifically, the amount of bailout funds is assumed to be an equity shock to the bank, which is absorbed by the inclusion of bank-time fixed effects, while the coefficient of interest measures post-program lending or default between branches located in areas with high and low pre-program defaults, conditional on the overall bailout amount received by the bank. Hence, the bank-district estimates can shed light on the within-bank reallocation of credit and changes in default after the program while controlling for unobserved contemporaneous shocks to the credit market.

We present results from the bank-district level regressions in Online Appendix E, Table OA.V. Similar to the main results, the estimates document a strong reallocation of post-program credit toward branches with lower defaults before the bailout as well as higher default in branches located in areas with greater program exposure. Taken together, these results suggest that our estimates are unlikely to be biased due to unobserved shocks to the credit market.

5.1.4 Plausibility of the exclusion restriction. We next present a number of tests to assess the exclusion restriction for the instrumental variable used in our analysis. The exclusion restriction requires that the instrument—the interaction between pre-program drought years and the share of households below the two-hectare collateral cutoff—affect outcomes only through its impact on program exposure. While this assumption cannot be tested directly, we present several tests to assess its plausibility.

The first potential challenge to the exclusion restriction is the possibility that the instrument may affect outcomes directly. To address this concern,

³⁵ This derivation, which we document in Online Appendix D, yields the following estimating equation:

$$\ln(C_{ijt}) = \lambda_{it} + \lambda_{bt} + \gamma E_{ijt} + \chi_{ijt} \quad (7)$$

where λ_{it} are district time fixed effects, λ_{bt} are bank-time fixed effects, E_{ijt} is the share of nonperforming agricultural loans at the branches of bank j located in district i , and χ_{ijt} is a stochastic error term.

Table 7, Panel A.1, tests for a correlation between the instrument and the main outcomes of interest in the pre-program period. To perform this test, we restrict the sample to a panel covering the period between 2001 and 2007 and estimate the conditional correlation between the instrument and our outcomes of interest, where the component of the instrument measuring the time series of weather shocks is lagged to capture a five-year window before each year $t \in \{2001, \dots, 2007\}$. The results show that there is no correlation between the instrument and any of our main outcomes of interest over this time period. This suggests that, while each component of the instrument may affect outcomes individually, the interaction between weather shocks and the two-hectare collateral cutoff indeed only becomes relevant once the program is announced.³⁶

The second challenge to the identification strategy is the possibility that cross-sectional variation in the instrument might be correlated with district characteristics at the time of the bailout, in a way that affects outcomes in the post-program period for reasons unrelated to the program. To address this concern, we first test for a cross-sectional correlation between the instrument and observable district characteristics at the time of the program announcement and find no evidence that this is the case. We then additionally augment our preferred specification with a set of interaction terms between district characteristics c_{ij} and the post-program dummy, to ensure that our results are not driven by a differential effect of the instrument on districts with different characteristics at the time of the program announcement. In addition to interactions with a district's total population, rural share, agricultural productivity at the time of the program announcement, we also include an interaction term between the post-program dummy and the share of landholdings larger than four hectares. This effectively restricts identification of the program impact to a band of ± 2 hectares of the cutoff, thus controlling for the possibility that the response to weather shocks differs in districts with a significant share of large landholdings. In Table 7, Panel B.1, we first report OLS estimates that control for interaction terms between district characteristics and the post-program dummy. In Table 7, Panel B.2, we report the parallel 2SLS estimates. Both sets of results are similar to the baseline estimates, which confirms that the results are not biased due to heterogeneity of the treatment response or a heterogeneous impact of the instrument by initial district characteristics.

Finally, since our instrument relies on the number of drought years experienced by a district ahead of the program, one may be concerned that the estimates could be biased due to persistence in local weather shocks. That is, if districts that were more likely to experience a drought in the years prior to the bailout were also more likely to experience a drought after the program, the

³⁶ This is also consistent with placebo tests in Table 7 that estimate treatment effects for arbitrary program dates and find no significant first-stage or treatment effects before the ADWDRS program announcement in 2008.

Table 7
Robustness: Exclusion restriction and endogeneity of treatment response

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Log(loans)	Log(amount)	NPL(loans)	NPL(amount)	Log(productivity)	Log(wage)	Log(consumption)
Panel A.1: Effect of the instrument on pre-program outcomes, $t \in \{2001, \dots, 2008\}$							
Drought years $_{i,pre}$ *Below cutoff	0.043 [0.041]	0.026 [0.048]	-0.035 [0.061]	0.033 [0.034]	0.001 [0.004]	0.132 [0.121]	0.077 [0.049]
Observations	2,487	2,487	400	757	2,386	316	1,104
R ²	0.666	0.870	0.188	0.289	0.400	0.120	0.968
Panel B.1: OLS treatment effects controlling for interactions with district characteristics $c_{ij}^j \cdot \text{post}_{2008}$, $t \in \{2001, \dots, 2012\}$							
Bailout share*post	-0.027* [0.015]	-0.030** [0.014]	0.069*** [0.021]	0.066*** [0.023]	-0.001 [0.002]	-0.023 [0.034]	0.018* [0.010]
Observations	4,291	4,291	2,535	2,508	2,581	420	1,228
R ²	0.765	0.936	0.238	0.216	0.402	0.337	0.948
Panel B.2: 2SLS treatment effects controlling for interactions with district characteristics $c_{ij}^j \cdot \text{post}_{2008}$, $t \in \{2001, \dots, 2012\}$							
Bailout share	0.019 [0.069]	-0.285*** [0.092]	0.356 [0.299]	0.754** [0.375]	0.005 [0.011]	-0.731 [0.986]	-0.868 [0.880]
Observations	3,990	3,990	2,361	2,334	2,449	405	1,157
R ²	0.761	0.911	0.168	0.021	0.402	0.104	0.212
Panel B.3: 2SLS treatment effects controlling for drought years $_{pre}$ *post interactions, $t \in \{2001, \dots, 2012\}$							
Bailout share*post	-0.034 [0.039]	-0.261*** [0.051]	0.287* [0.160]	0.521*** [0.176]	0.004 [0.009]	-0.203 [0.329]	-0.594 [0.384]
Observations	4,096	4,096	2,427	2,398	2,482	408	1,172
R ²	0.759	0.913	0.181	0.172	0.378	0.178	0.555
District FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
State*Electoral cycle FE	Y	Y	Y	Y	Y	Y	Y

This table reports additional robustness checks. Panel A.1 restricts the data set to the years 2001–2007 and tests for a correlation between the instrument and pre-program outcomes. Panel B.1 augments the baseline OLS specification with interaction terms between district characteristics c_{ij} and the post-program dummy. These district characteristics include the total population, rural population share, per capita income, agricultural productivity, and the share of landholdings larger than four hectares. Panel B.2 reports 2SLS results controlling for interaction terms between district characteristics and the post-program dummy. Panel B.3 presents 2SLS results controlling for interactions between the number of pre-program drought years and the post-program dummy. Standard errors, in brackets, are heteroscedasticity-robust and clustered at the district level. ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

treatment effects could be driven by these persistent weather shocks rather than the impact of the bailout. To address this potential confound, Table 7, Panel B.3, presents estimates that control for interactions between the number of pre-program drought years and the post-program dummy. The point estimates are nearly identical to those from the baseline specification, suggesting that the results are not biased as a result of persistence in local weather shocks.

Taken together, these results suggest that the exclusion restriction is unlikely to be violated by either a direct effect of the instrument on program outcomes, persistent weather shocks, or a correlation with omitted district characteristics that might affect the treatment response.

5.2 External validity

Two features of our setting are relevant to the external validity of our study. First, as in many emerging markets, rural credit in India is provided primarily by state-owned banks. This could affect our results if there had been differences in the incentive environment faced by public- and private-sector banks ahead of the bailout, or if responses to the bailout differ between private and state-owned banks. The first concern is minimized by the fact that public and private banks in India face identical regulation and subsidy policies for agricultural credit, so that the nature of the incentive distortion prior to the bailout would have been the same.³⁷ However, responses to the bailout could differ by bank ownership. Under the assumption that private banks respond more strongly to market incentives (La Porta, Lopez-De-Silanes, and Shleifer 2002), our estimates on post-program lending and real economic activity should be interpreted as lower bound estimates, while the effects of the bailout on post-program moral hazard should be interpreted as an upper bound—assuming that borrowers believe the consequences of defaulting on a private-sector bank to be more severe than those of defaulting on a government-owned bank. Second, there are of course differences in the legal and institutional environment across countries, so that while our study sheds light on the impact of debt relief in an important setting—one of the world's largest emerging market economies—there may of course be some limitations in the extent to which our results apply to debt relief initiatives in settings with more developed credit markets and legal institutions.

6. Conclusion

The world over, governments have routinely intervened in credit markets in an effort to stimulate economic activity. While credit market stimulus programs are extremely common, they are often thought to have adverse implications for credit allocation and repayment discipline, without generating sufficiently large

³⁷ As noted above, our regressions control for the state-level electoral cycle to account for political cycles in state bank lending, which have been documented by Cole 2009, Carvalho (2016), and others.

offsetting effects on real economic activity. There exists, however, surprisingly little empirical evidence to evaluate these claims.

In this paper, we have used a natural experiment arising from one of the largest borrower bailouts in history to estimate the causal effect of a large credit market stimulus on credit market outcomes and the real economy, and make two main contributions.

First, we examine the impact of the bailout on post-program credit allocation and show that the program led to a significant reallocation of bank lending away from districts with greater bailout exposure—possibly due to the anticipation of the moral hazard costs of the bailout. While this pattern of post-program lending has adverse effects for economic activity in regions with high bailout exposure, it unambiguously improves the efficiency of credit allocation, as new lending is reallocated toward observably less risky and more productive borrowers.

Second, we find that the moral hazard costs of the bailout are substantial, and not offset by a stimulating effect on the real economy. We document a sharp increase in defaults, concentrated among borrowers that were not in default at the time the program came into effect, and in districts with a larger share of bailout beneficiaries, as well as higher average amounts of debt relief per beneficiary. This result is consistent with peer effects in default and suggests that the bailout had a significant negative impact on the repayment behavior of nonbeneficiaries. While it is possible that the bailout had a positive effect on investment incentives in regions with high program exposure, its potential effect on economic activity is entirely offset by the reallocation of credit away from high bailout regions, so that debt relief has no effect on real economic activity.

These results highlight the difficulty of designing debt relief programs in a way that minimizes moral hazard and ensures the transmission of the credit market stimulus to the real economy. In particular, our findings underscore the importance of taking into account the impact of debt relief on post-program credit supply. The credit reallocation effect we find is large enough to prevent any transmission of the program to the real economy through the incentive or collateral channel, and is likely exacerbated by two features of the program. First, the bailout covered primarily term loans with short maturity, which were fully written off and eliminated from banks' balance sheets as soon as the program came into effect. Hence, banks were free to immediately reallocate credit away from regions with high bailout exposure to avoid the predictable moral hazard issues generated by the program. Second, eligibility for the bailout was conditional on default status and land collateral and therefore not orthogonal to borrower characteristics. Moreover, the program made debt relief mandatory for borrowers below the collateral threshold, thus treating willful defaulters and genuinely distressed borrowers alike. This aggravated moral hazard among borrowers who could have repaid but were bailed out, and among borrowers who did not qualify for debt relief because they had remained current on their loan payments throughout.

Finally, our results provide evidence that the moral hazard effects of the bailout are likely to be aggravated by political economy factors. We show that the relationship between loan defaults and the electoral cycle—which is present in the data in normal times—is magnified after the bailout. This suggests that the costs of a credit market stimulus are likely to be especially severe in environments with weak legal institutions and a history of politically motivated credit market interventions.

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